Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Subject: Selection Functionals Paper - My Response to Your Latest Comments

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 24, 2026 at 3:20 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>
Cc: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>

Hi Ezequiel and Sara,

Thank you both. Revision 3 is on SSRN (https://dx.doi.org/10.2139/ssrn.6922758). It makes the changes below and no others; every section and result number is unchanged, and the paper's only attribution remains the Acknowledgments sentence. I address Ezequiel's comments first followed by Sara's...

Ezequiel,

Amendment 1. The step is wrong for the reason you give. The contraction factor of the Schur recursion at its fixed point is $1 - 2k_{\mathrm{eff}}\,\delta t$, measured as 0.9378, 0.9961, 0.9990 and 0.9995 at $M = 50$, 800, 3200 and 6400, and the relative deviation reaches 10^{-6} after 5189 steps at $M = 6400$ and not within the grid below, as you found; so the transient is a fixed fraction of the horizon and the bounded-layer claim fails. The recursion from the last index backward gives the forward innovation variances: the Schur complement of the future block is the precision of the marginal law of the past, whose last entry is the conditional precision of ξ_m given ξ_{m-1}, and the recursion's values agree with $\mathrm{Var}(\xi_m \mid \xi_{m-1})$ computed from the covariance to 10^{-16}. The recursion is also monotone and contracting from its boundary value, which lies within $O(\delta t)$ of the fixed point, so every innovation variance is $\sigma_{\mathrm{eff}}^2\delta t(1 + O(\delta t))$; the proof claimed a bounded transient, the claim is false, and this $O(\delta t)$ bound would have sufficed. The revision replaces the proof by the trace argument, which needs no statement about the recursion. In the increment coordinates $Y_m = \xi_{m+1} - \xi_m$ the selected precision is $(1/\delta t)(uI + E)$ by Proposition 3.5, with $u = a_\Pi/\sigma_\Pi^2 + \lambda a_\nu/\sigma_\nu^2$ and $E = \delta t\kappa \mathbb{1}\mathbb{1}^{\mathsf{T}} + \delta t^2\kappa_2 L^{\mathsf{T}}L$ positive semidefinite, where $\kappa = k_\Pi/\sigma_\Pi^2 + \lambda k_\nu/\sigma_\nu^2$, $\kappa_2 = k_\Pi^2/\sigma_\Pi^2 + \lambda k_\nu^2/\sigma_\nu^2$ and L is the strictly lower-triangular matrix of ones. So $C_M = \delta t(uI + E)^{-1} \le (\delta t/u)I$, which gives $\mathrm{Var}_Q(V_M) = 2\,\mathrm{tr}\,C_M^2 \le 2T\delta t/u^2$, and $\mathrm{tr}\,E = \kappa T + \kappa_2 T(T - \delta t)/2$ gives $T/u - (\delta t/u^2)(\kappa T + \kappa_2 T^2/2) \le E_Q[V_M] \le T/u$; $u \to 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2$, and Chebyshev's inequality gives $V_M \to \sigma_\infty^2 T$ in $Q_{\nu,M}$-probability. On Example I, $\mathrm{tr}\,C_M$ and $2\,\mathrm{tr}\,C_M^2$ reproduce your six figures each to six decimals, the two bounds hold at every M, and the closed form $\delta t(uI + E)^{-1}$ agrees with $DK^{-1}D^{\mathsf{T}}$ to 10^{-16}. The parenthetical now reads: if the target's quadratic variation converges in probability to a random limit V, then $\liminf d_{TV} \ge 1 - P(V = \sigma_\infty^2 T)$, so $d_{TV} \to 1$ whenever the law of V has no atom at $\sigma_\infty^2 T$.

Amendment 2. The abstract now reads $1/\sigma_{\mathrm{eff}}^2 = 1/\sigma_\Pi^2 + \lambda/\sigma_\nu^2 + O(\delta t)$, "to leading order in the grid step". The same formula stood without the remainder in the introduction's paragraph on the weight and in contribution (C1), and both carry it now.

Verification. Twenty-four of your twenty-six re-derived figures agree with the archived numerics to every printed digit; your 20.3342 is the asymptotic variance at $k = 2.76$ exactly, and the archive's 20.3166 is at the minimizer 2.7618. The two interior increment variances, 0.01010 and 0.02012 in your reply, are 0.010115 and 0.020131 in the archive; the paper prints 0.0101 and 0.0201, which both give, and the fourth significant digit depends on the rows averaged. On the Monte Carlo: your third run, 3.817 ± 0.127, and mine, 4.025 ± 0.052, differ by 1.5 standard errors of the difference, and a random 1,000-replication subsample of my run lands at or below 3.817 with probability 0.086, so the two runs are compatible. The 3.44 lies 3.6 standard errors from mine and 1.9 from your third run (taking your first-run standard error as $0.127\cdot\sqrt{(1000/700)}$), so the statement that the three gaps exceed sampling noise holds for the gap between 3.44 and 4.02 only. Your maximum 5.79 over 4,000 batches lies below the smallest maximum over 4,000 batches resampled from my run, 7.29 (median 7.95), so the two implementations differ in the upper tail of the batch statistic, in the direction opposite to the 7.53. Both conclusions of Finding 7 hold in all three runs.

The display of K_Π in Example 7.4 is unchanged; the typeset page is the deposited text. I have carried your note on Open Problem (v) to the memory-rank paper's record.

Sara,

First point. The comparison paragraph said the novelty claim rests on the distinction between a product and a ratio of densities. It now says that products of densities are established, the product of experts of Hinton (2002), and that the claim rests on the exact product identity (Proposition 3.5), its finite-grid implementation, and the consequences derived here: Corollary 3.6, the three consequences of Section 3.4, the refinement mechanism of Section 9.4 and the design rule of Section 7.5.

Second point. The manuscript defines the intensity as the two-sided integral, $\sigma\sigma^{\intercal} = \int_{\mathbb{R}} E[f(t)f(0)^{\intercal}]dt$; my letter wrote the integral without its limits. Section 8.3 now also writes it as $\int_0^{\infty}(R_f(t) + R_f(t)^{\intercal})dt$ with $R_f(-t) = R_f(t)^{\intercal}$, and states that the one-sided integral γC alone is neither symmetric nor positive semidefinite in general.

Third point. "The estimator has no limit" in Section 11 and "the estimator does not survive refinement" in Section 1 now read "the fixed-proposal estimator has no nondegenerate limit"; Section 9.4 already said "from Π_M at any fixed $\lambda > 0$". Section 11 adds that a joint limit in which N grows with M at least as fast as $1 + \chi^2(Q_{v,M}\|\Pi_M)$, or a sequence of proposals adapted to the selected law, is not excluded and is not studied here.

Ben

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On Wed, Sep 23, 2026 at 4:08 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Ben, hi again;
>
> My assessment of this paper is favourable in substance, subject to two focused amendments set out at the end.
>
> **Your correction of my Finding 2 is right.** My equation (1) presented $1/\sigma_{eff}^2 = 1/\sigma_{\Pi}^2 + \lambda/\sigma_v^2$ as exact, and it is not.
> From the interior-row matching, $v - 2u = (1-a_{\Pi})^2/\sigma_{\Pi}^2 + \lambda(1-a_v)^2/\sigma_v^2$, which is exactly $\delta t^2(k_{\Pi}^2/\sigma_{\Pi}^2 + \lambda k_v^2/\sigma_v^2)$, so the exact relation is the one in k^2/σ^2
> and mine is its leading order.
> On Example I the exact relation holds with residual $2.4\cdot10^{-13}$ while mine is off by 0.0033, and on refining δt from 0.02 to 0.0002 the ratio of that discrepancy to
> δt sits at 0.165, 0.163, 0.162, 0.162 (first order, not rounding). Corollary 3.6(ii) states both with the $O(\delta t)$ where it belongs.
> My report should have done the same.
>
> **Verification of the new material.** I re-derived it from the definitions rather than checking your arithmetic against mine.
>
> - **Corollary 3.6 on Example I.** $\sigma_{eff}^2 = 0.499177$, $k_{eff} = 1.5798$, and mean interior increment variance 0.01010 under
>   Q against 0.02012 under Π. $\mathrm{Var}_{\Pi}(\xi_M) = 0.438071$ and $\chi^2(Q\|\Pi) = 1372.4$, both exact.
>   The asymptotic variances reproduce: 30.8925 at k = 2, 4.9987 at k = 12.39, 20.3342 at k = 2.76.
> - **Calibration.** The closed form gives $k^* = 2.7618$ with the sum over $m = 0,\ldots,M-1$, and the kinetic sum alone minimises at 1.000.
> - **Lemma 8.1.** $\lambda c_2\delta t\|\Sigma_{\Pi}\| = 0.4519$ at $\lambda = 3/4$ and 0.6025 at $\lambda = 1$, with $\|\Sigma_{\Pi}\| = 10.04$.
> - **Example 7.4.** Least eigenvalues 0.3387 and 1.2703; at the triple (3,5,7) the direct coefficient is −0.008588 against a composed 0.305127,
>   both exact, with gap 0.314; none of the 56 triples composes and the smallest gap is 0.011, at (6,7,8). And $K_{\Pi} = K_{target} - \lambda K_v$ identically.
>   The sentence you added (that zero bias is attained by presupposing the target law) is the right delimitation of what my construction shows,
>   and neither review made it.
> - **Example II.** I rebuilt the double-well benchmark by transfer-matrix quadrature on a mesh of my own.
>   With the divergence term $F^* = -0.547760$, $\chi^2 = 162.1$, $\sigma^2 = 44.31$; without it, −0.637163, 169.3, 26.60; the proposal's value −0.561929;
>   the term shifts the estimand by 0.089 and reverses the sign of the displacement. All six figures reproduce to every printed digit.
>   That answers Finding 6 completely, and the sign reversal is a stronger result than the finding asked for.
>
> One caution for anyone reading the deposit as extracted text rather than as a typeset page: the display of K_{Π} in Example 7.4 comes out of the
> PDF with ε inside the $1/\delta t$ factor, and under that reading the proposal has least eigenvalue −15.42 and is not a probability measure.
> The typeset page has ε outside and everything checks. I mention it because I nearly reported it as an error.